

ASTREA

AI-Powered Personal Reminder Butler

TECHNICAL DOCUMENTATION

Overview

Astrea is a cross-platform AI reminder assistant. Users speak or type naturally, and Astrea understands intent, creates structured reminders, syncs across devices in real-time, and sends push notifications when reminders are due.

Tech Stack

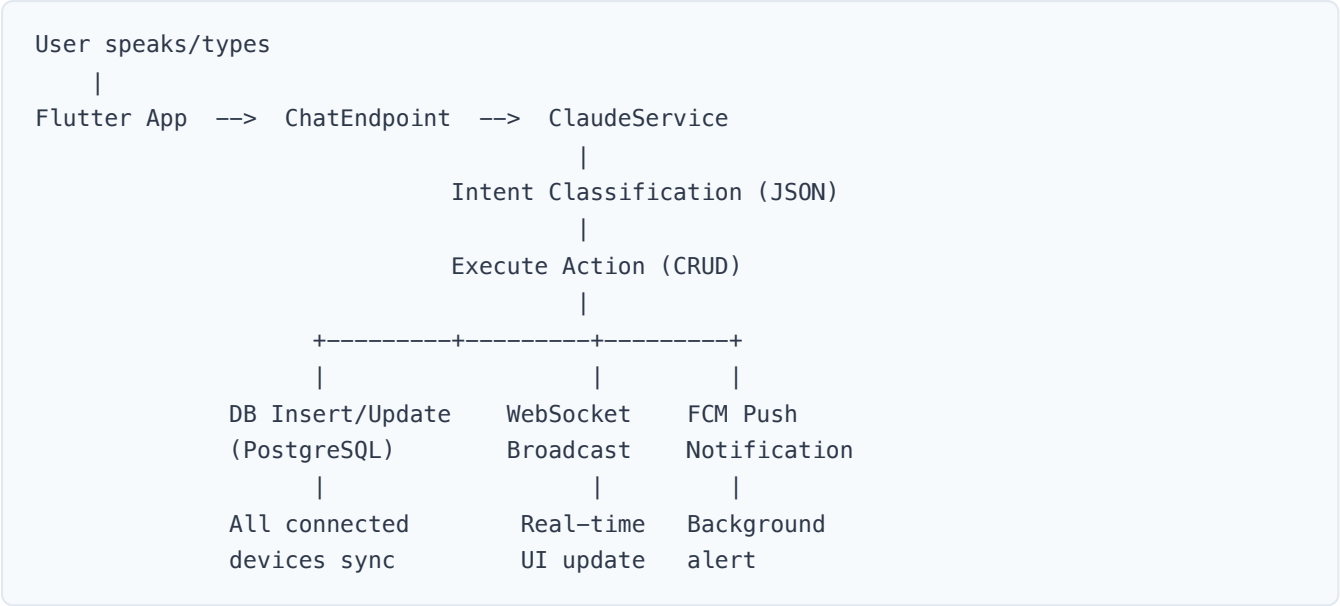
Layer	Technology	Purpose
Frontend	Flutter 3.32 + Dart 3.8	Cross-platform mobile app
State Management	Riverpod	Reactive state with providers
Backend	Serverpod 3.2.3	Type-safe Dart backend framework
Database	PostgreSQL + pgvector	Storage + vector similarity search
AI – Chat	Claude Sonnet (Anthropic API)	Intent classification + NLP
AI – Search	Gemini text-embedding-004	768-dim embeddings for semantic search
Push Notifications	Firebase Cloud Messaging	Background sync + alerts
Local Notifications	awesome_notifications	Scheduled alerts with snooze/complete
Voice Input	speech_to_text	Device-native STT (no cloud API)
Subscriptions	RevenueCat	In-app purchases + entitlements
Email	MailerSend HTTP API	Email verification codes
Hosting	Railway	Docker deployment + managed Postgres

Architecture

Monorepo Structure

```
astrea/  
  astrea_server/  Serverpod backend (endpoints, services, models)  
  astrea_client/  Auto-generated type-safe client SDK  
  astrea_flutter/  Flutter mobile app (Riverpod + UI)
```

Data Flow



AI Intent Classification

Claude Sonnet receives the user message with conversation history and current reminders as context. It returns structured JSON with an `intent` and `action` parameters. No regex or keyword matching is used — all understanding is delegated to the LLM.

Intent	Trigger Examples
<code>create_reminder</code>	"Remind me to call mom tomorrow at 3pm"
<code>update_reminder</code>	"Change my meeting to 4pm"
<code>complete_reminder</code>	"Done with groceries"
<code>delete_reminder</code>	"Delete the gym reminder"
<code>snooze_reminder</code>	"Snooze it for 30 minutes"
<code>list_reminders</code>	"What do I have this week?"
<code>question</code>	"When is my dentist appointment?"
<code>chat</code>	"Hello", "Thanks"

Semantic Search (RAG)

Each reminder is embedded into a 768-dimensional vector using `Gemini text-embedding-004`. Vectors are stored in PostgreSQL via the `pgvector` extension. When users ask questions like "what do I have next week?", the query is embedded and matched against stored reminders using cosine similarity — finding results by meaning, not keywords.

Real-Time Sync

Serverpod's WebSocket streaming pushes reminder changes to all connected clients instantly. When a reminder is created, updated, or completed on one device, a `ReminderSyncBroadcaster` sends the event through WebSockets. FCM push notifications handle the case when the app is backgrounded.

RevenueCat Integration

Configuration

Key	Value
SDK	<code>purchases_flutter: ^9.11.0</code> + <code>purchases_ui_flutter: ^9.11.0</code>
Entitlement	<code>Astrea Pro</code>
Products	<code>monthly</code> , <code>yearly</code>
Paywall	RevenueCat remote paywall (RevenueCatUI)

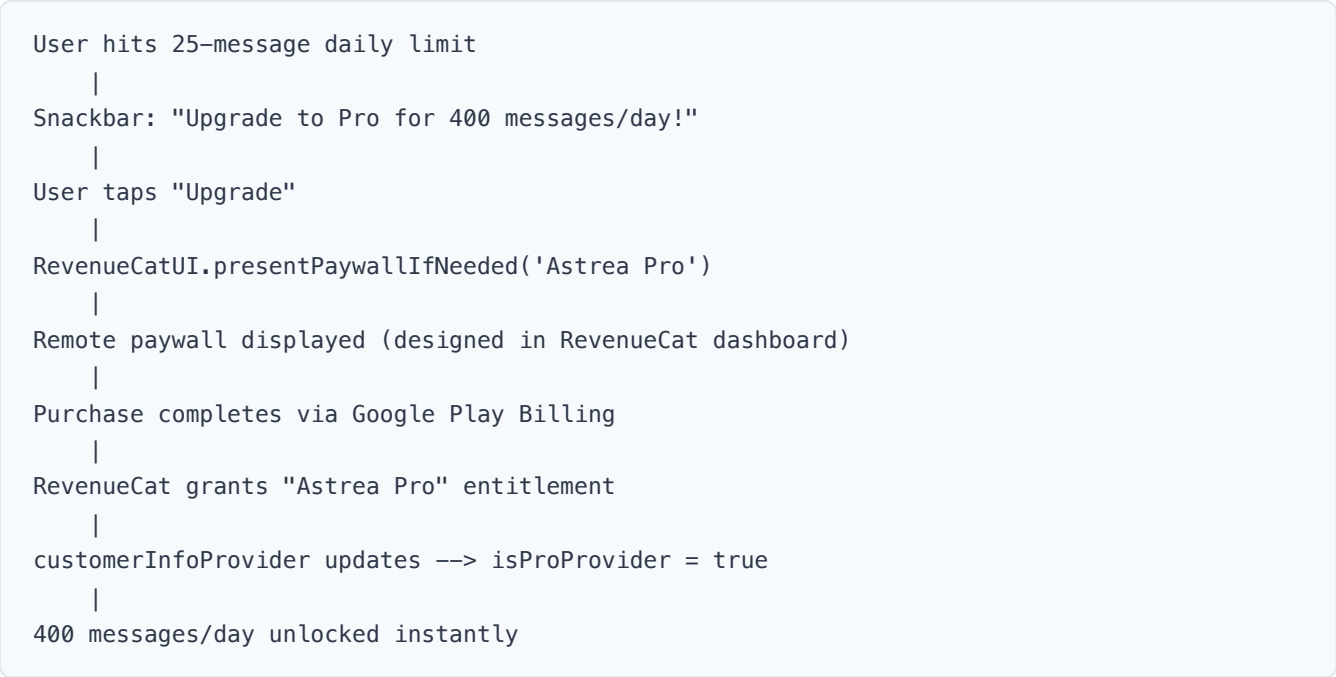
Monetization Model

Free Plan • 25 AI messages per day • Unlimited reminder management • Cross-device sync • Push notifications	Astrea Pro • 400 AI messages per day • All Free features included • Monthly or Yearly subscription • Manage via Customer Center
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Implementation

Component	File	What It Does
SDK Init	<code>main.dart</code>	Configures RevenueCat with API key on app startup
Subscription State	<code>subscription_provider.dart</code>	Riverpod providers: <code>isProProvider</code> , <code>customerInfoProvider</code> , <code>dailyMessageProvider</code>
User Identity	<code>auth_wrapper.dart</code>	<code>Purchases.logIn()</code> on auth, <code>Purchases.logOut()</code> on sign-out
Message Gating	<code>chat_provider.dart</code>	Checks daily limit before each AI call; throws <code>MessageLimitExceededException</code>
Upgrade Prompt	<code>chat_input.dart</code>	Catches limit exception, shows snackbar with "Upgrade" → paywall
Settings	<code>settings_page.dart</code>	Shows Pro/Free status; opens Customer Center or paywall
Android	<code>MainActivity.kt</code>	<code>FlutterFragmentActivity</code> (required for RevenueCat UI)

Purchase Flow



Daily Message Counter

The `DailyMessageNotifier` persists message count in `SharedPreferences`. It resets at midnight (comparing stored date vs current date). Pro users bypass the counter entirely. The counter is checked client-side before each API call, preventing unnecessary server requests when the limit is reached.

Deployment

Service	Platform	Details
Backend	Railway (Docker)	<code>https://astrea-production.up.railway.app</code>
Database	Railway (pgvector)	PostgreSQL with vector extension, internal networking
Android App	Firebase App Distribution	Release APK with production config
Secrets	Railway env vars	<code>SERVERPOD_PASSWORD_*</code> pattern

Testing

36 unit tests covering subscription logic and chat state management. **31 integration tests** on the server covering endpoints.

Test Suite	Tests	Covers
subscription_provider_test	17	Daily counter, limit enforcement, Pro bypass, midnight reset, remaining messages
chat_provider_test	19	ChatMessage model, notifier operations, dissolution, loading state
Server integration tests	31	Reminder CRUD, chat, user settings, sync endpoints